Story 3: Why Do Road Deaths Differ So Much Between States?

DATA 608 · Fall 2026

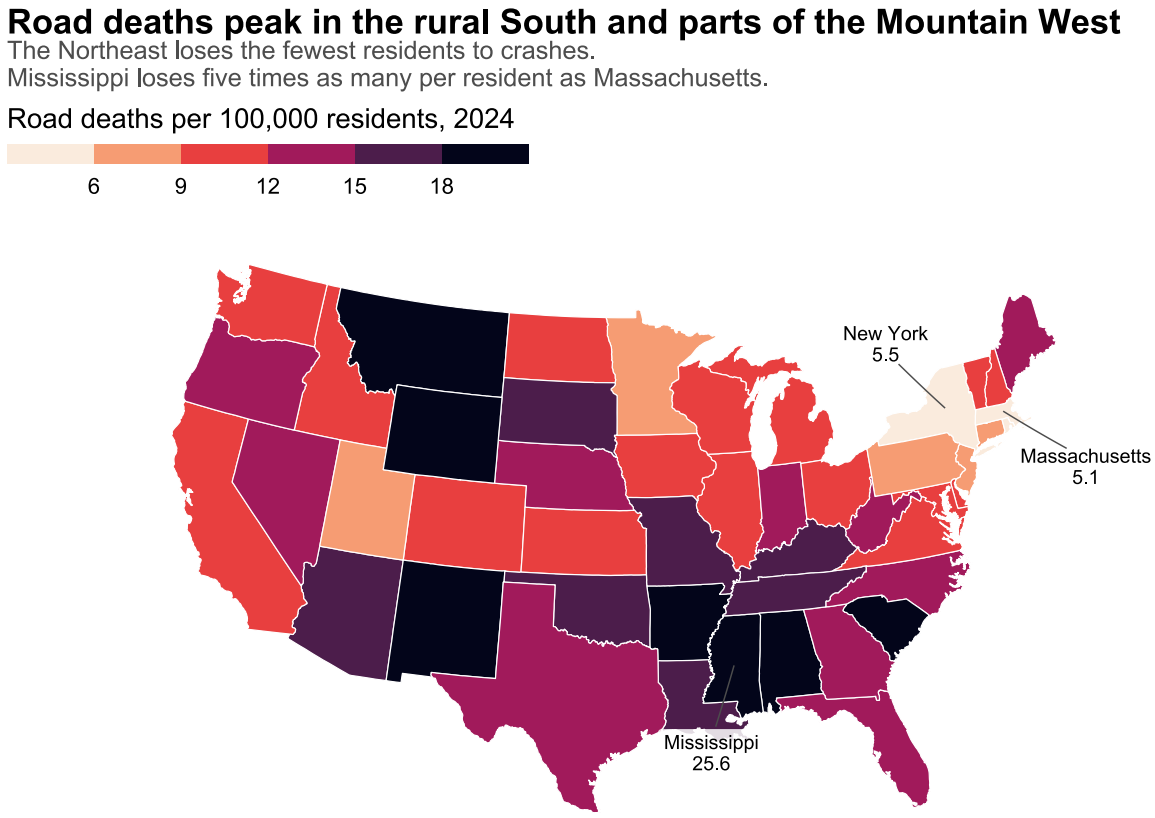
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Question: Why is someone in Mississippi about five times more likely to die on the road than someone in Rhode Island?

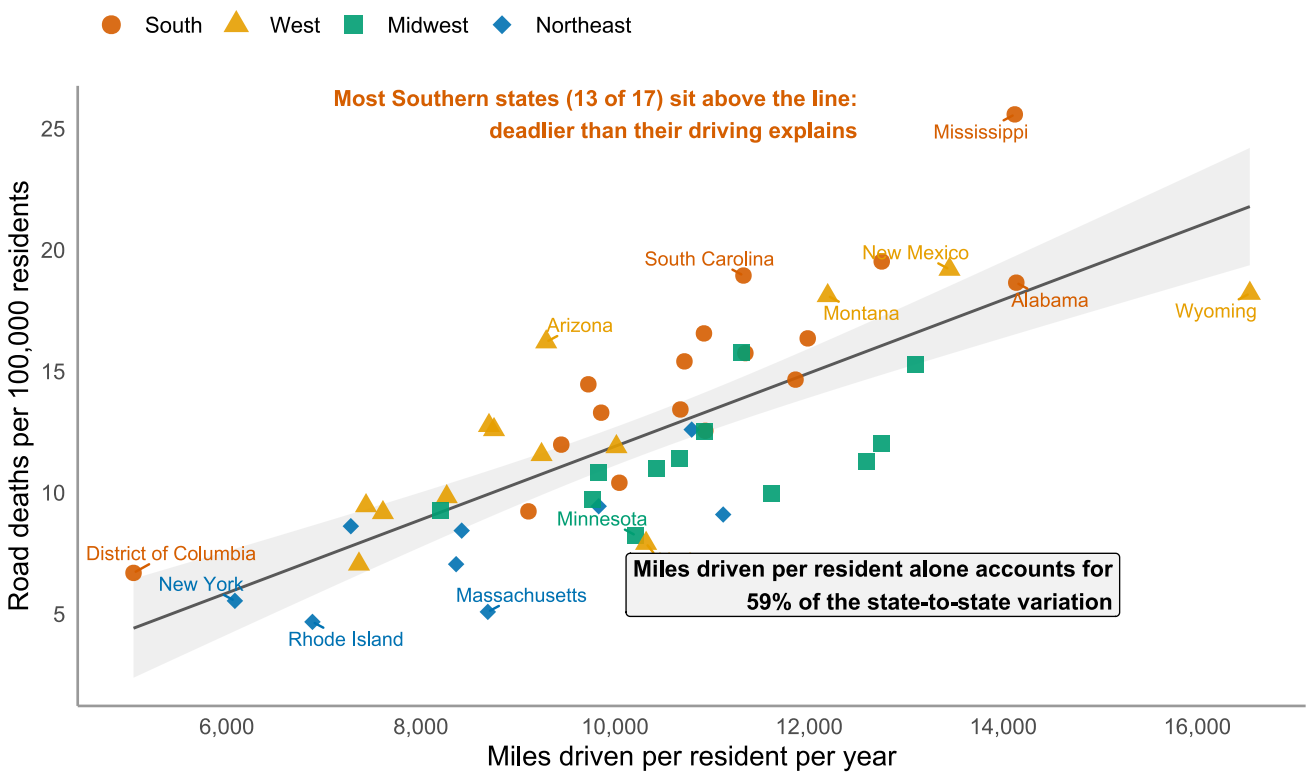
Short answer: Most of the gap tracks how much and where people drive. The South is deadlier even per mile, rural crash victims wait longer for help, and seat belt laws are the one factor a state can change directly.

The starting point is where road deaths are highest.



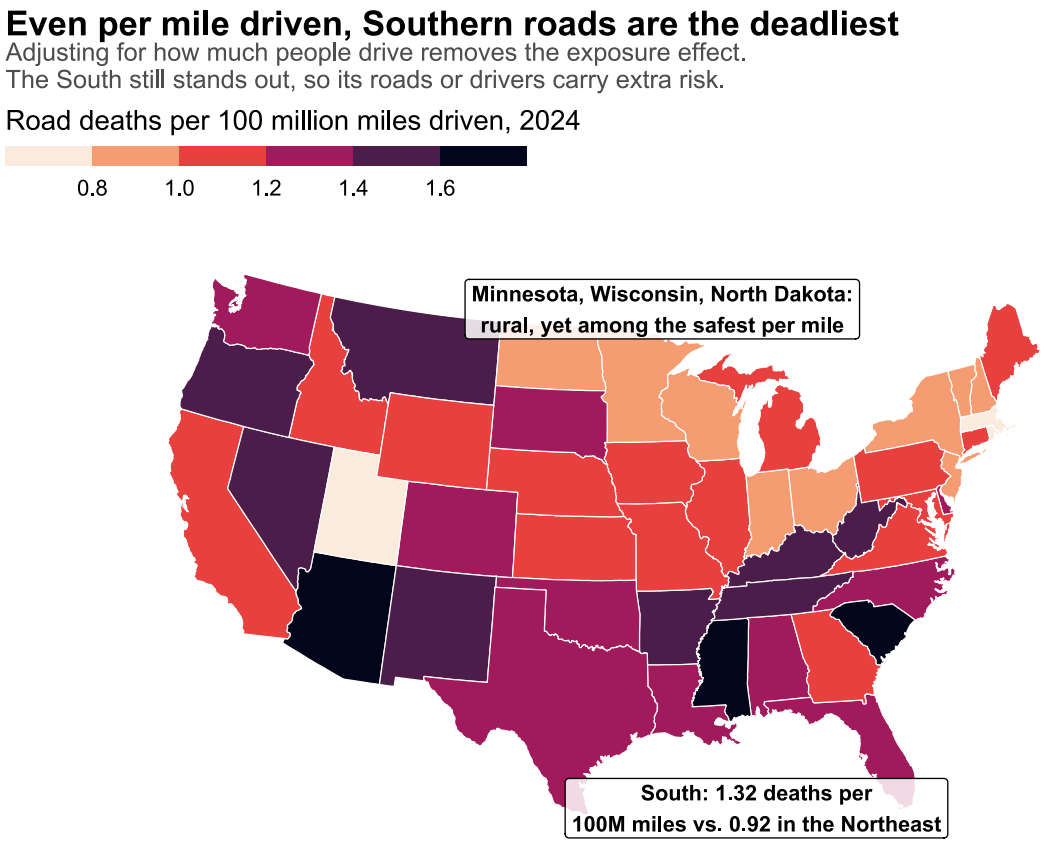
Source: IIHS Fatality Facts 2024. Alaska (9.5) and Hawaii (7.1) not shown.

The first explanation to test: do people in these states simply drive more?



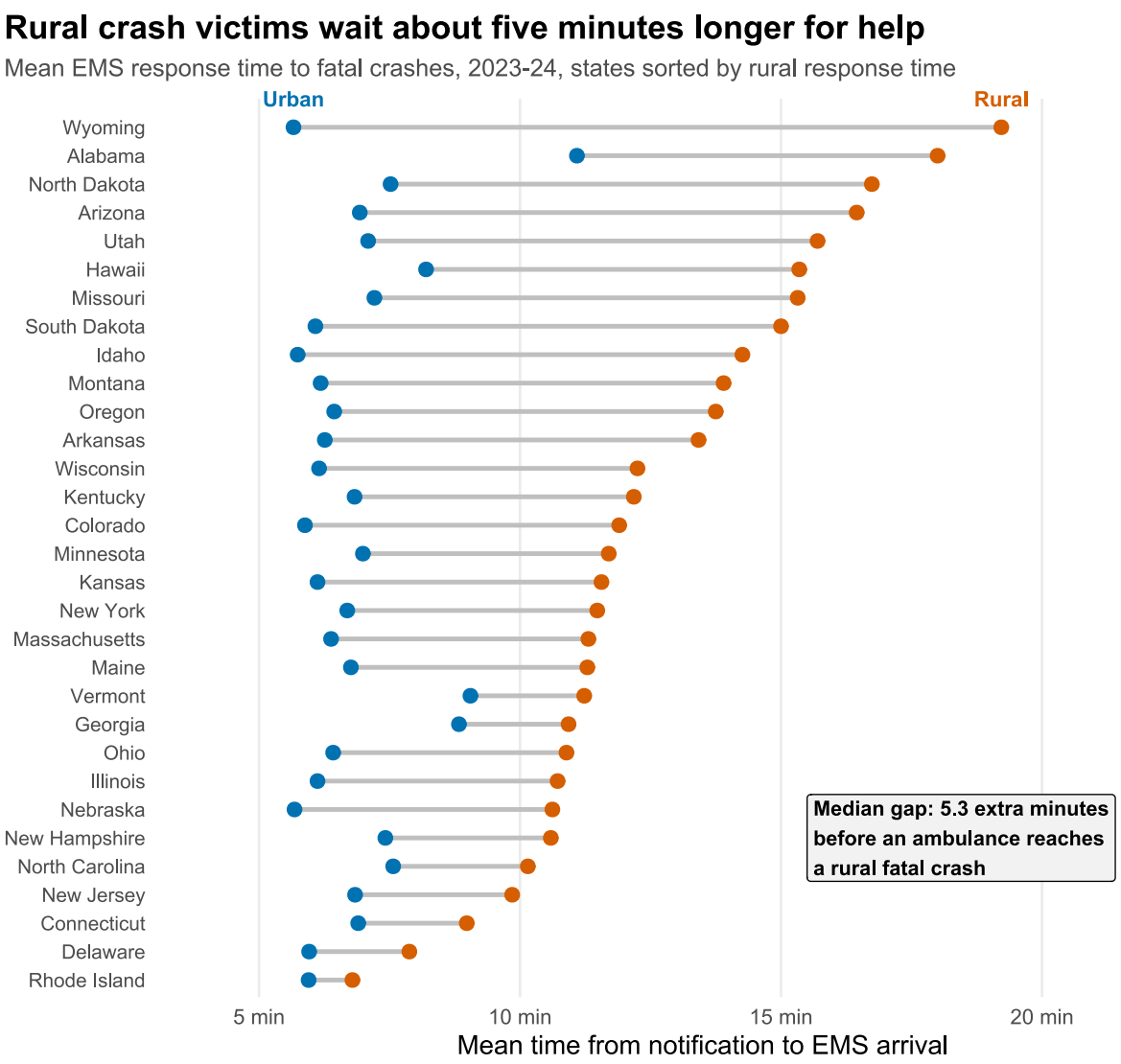
Sources: IIHS Fatality Facts 2024 (deaths, miles driven); Census regions.

Next: does the South still stand out once miles driven are accounted for?



Source: IIHS Fatality Facts 2024. Alaska (1.27) and Hawaii (0.96) not shown. Regional values are pooled deaths / pooled miles.

Rural roads may also matter after a crash, through how long it takes an ambulance to arrive.

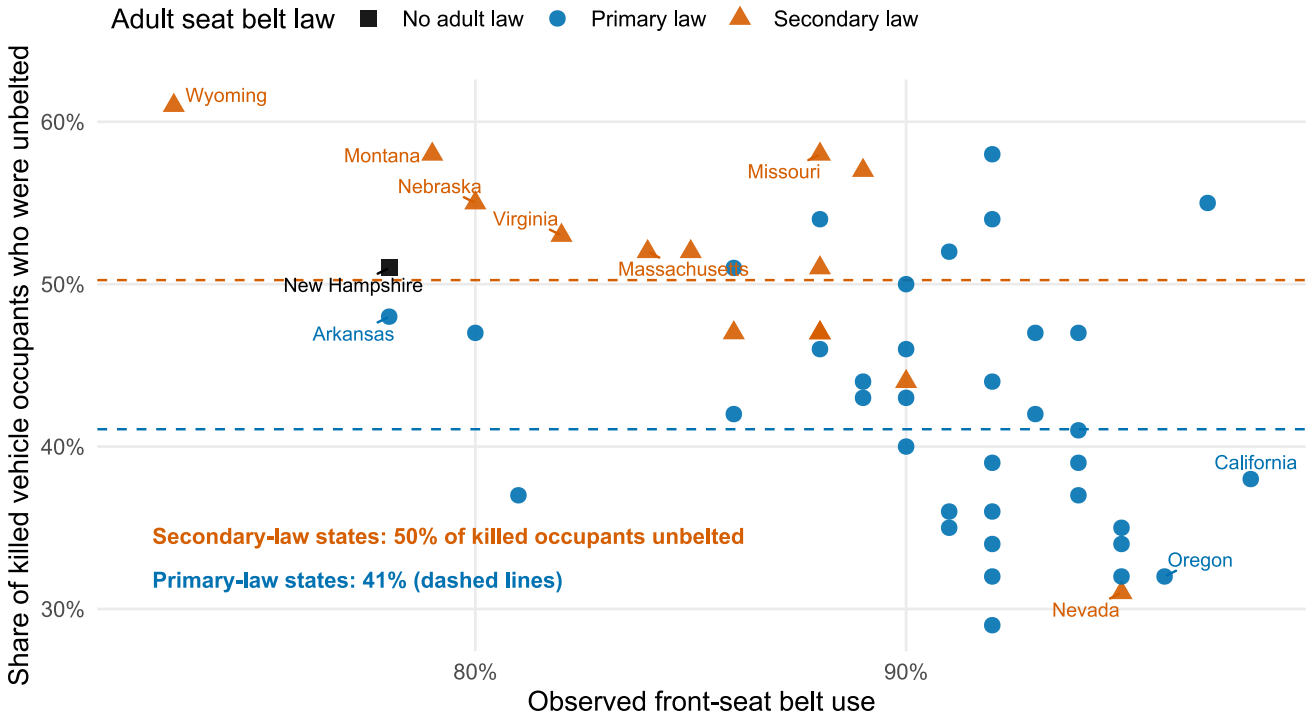


Source: FARS 2023-24 via course EMS summary. Only the 31 states reporting EMS times for at least 60% of fatal crashes are shown. CA, FL, TX, VA, NM, SC and others report too few to compare.
Times cover fatal crashes only, not the average crash response.

Driving habits, geography and ambulance distances are hard for a state to change. Seat belt laws are a direct policy lever, so the story ends there.

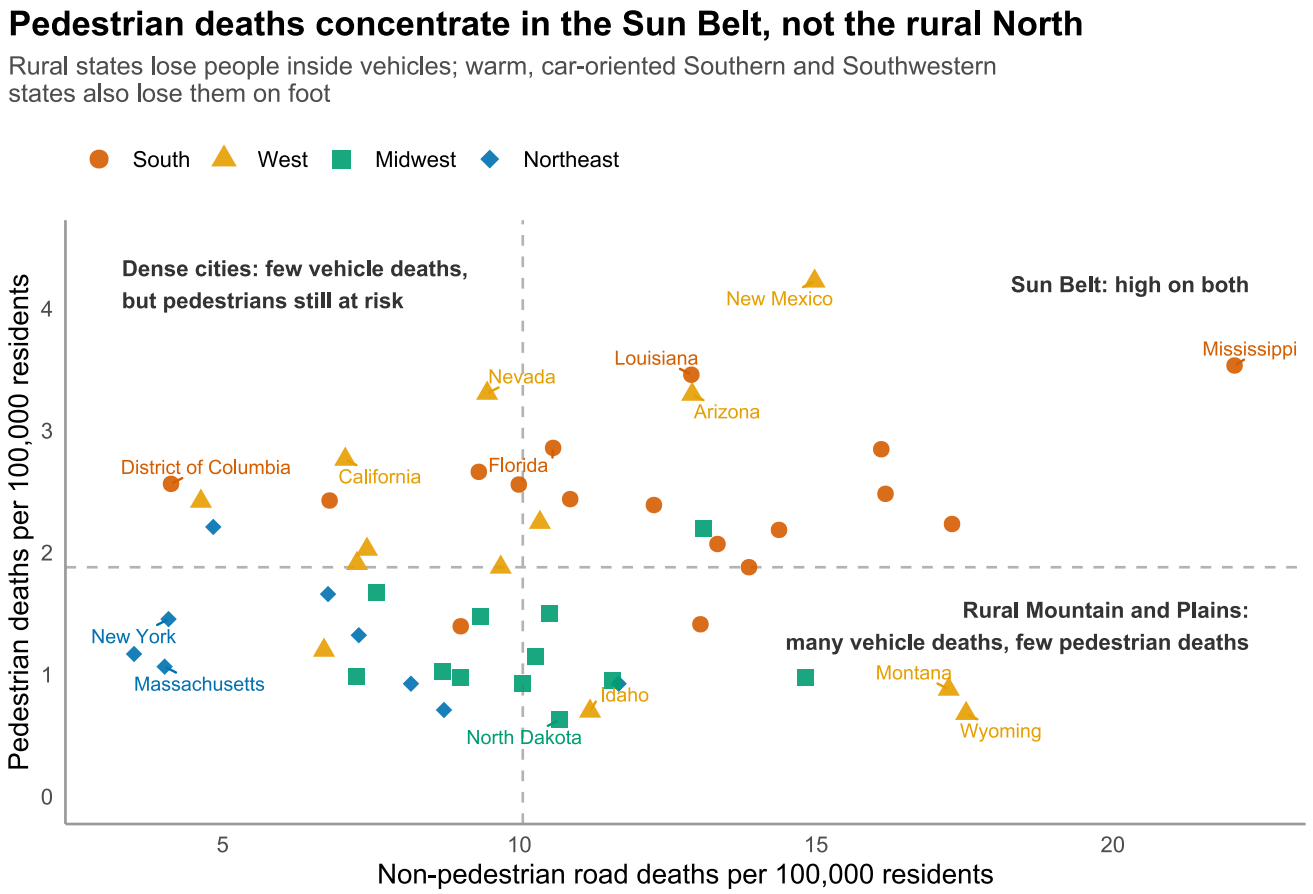
Weaker seat belt laws mean more unbelted people among the dead

Unlike miles driven or rural geography, belt laws are something a state can change directly.
Where police can't stop drivers for a belt violation alone, fewer people buckle up.



Source: IIHS Fatality Facts 2024; IIHS seat belt law table. Dashed lines are death-weighted averages by law type.
Primary law: police may stop a driver for a belt violation alone. Secondary law: a belt ticket only during another stop.

One more pattern: pedestrian deaths follow a different geography.



Source: IIHS Fatality Facts 2024. Dashed lines are state medians.
Rates are per resident, not share of deaths: DC's 38% pedestrian share reflects few vehicle deaths, not high pedestrian risk.

Limitations

- These are state-level associations, not proven causes. Heavy driving, rural roads and slow ambulances cluster in the same states, and state averages don't describe individual drivers.
- Causation may also run the other way: states with more deaths may face more pressure to pass primary belt laws.
- Small states (DC, Rhode Island, Vermont, each under 60 deaths) have noisy rates. Speeding and drunk driving were also checked but showed no clear link to death rates, so they are left out.

Data: IIHS Fatality Facts 2024 and the FARS 2023-24 EMS summary provided for the course. The CSVs are stored at https://github.com/tagensingh/Data608_FA2026_Story_3.